Telephony Provider Costing & Technical Implementation Analysis

Comparison of Airtel Corporate SIP Trunking vs. Exotel Cloud Telephony vs. Twilio Voice API for AI Calling Platform | 100 Calls/Day (12,000 Mins/Month)

Executive Summary

This report evaluates three PSTN telephony integration strategies to deliver AI voice streams directly into a customer's phone ear receiver without requiring any application on the customer's device. The evaluation models a standard operational load of **100 calls/day** across **30 days/month** with an average duration of **4 minutes per call** (Total volume: **3,000 calls / 12,000 call minutes per month**).

1. Monthly Cost Breakdown Comparison (in INR ■ & USD $)

Cost Parameter	Airtel Corporate SIP Trunk	Exotel Cloud Telephony	Twilio Voice API
Outbound Call Rate / Min	■0.30 / min ($0.0036)	■0.60 / min ($0.0071)	■1.52 / min ($0.0180)
Monthly Trunk / Rental Fee	■2,500 ($30.00)	■2,000 ($23.80)	■97 ($1.15)
Telephony Cost (12k Mins)	**■6,100 ($72.60)**	**■9,200 ($109.50)**	**■18,241 ($217.15)**
AI Pipeline (STT+LLM+TTS)	■10,634 ($126.60)	■10,634 ($126.60)	■10,634 ($126.60)
Total Monthly Cost (3k Calls)	**■16,734 INR ($199.20)**	**■19,834 INR ($236.10)**	**■28,875 INR ($343.75)**
Cost Per 4-Min Call	**■5.58 INR ($0.066)**	**■6.61 INR ($0.078)**	**■9.62 INR ($0.114)**
Effective Cost Per Min	**■1.39 INR / min**	**■1.65 INR / min**	**■2.40 INR / min**

Key Financial Takeaways:

- **Airtel Corporate SIP Trunking** is the most cost-effective solution for high-volume Indian operations, yielding a total cost of **■1.39/min** (~67% cheaper than Twilio).
- **Exotel** provides the best balance of low Indian call rates (**■1.65/min** total) with zero physical server hardware maintenance.
- **Twilio Voice API** is the most expensive for Indian calling (**■2.40/min** total) due to international USD-denominated tariffs, but offers global deployment simplicity.

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2. Technical Implementation Architecture for Each Option

Approach A: Airtel Corporate SIP Trunk (Self-Hosted FreeSWITCH / LiveKit SIP)

Architecture: Airtel provisions a Session Border Controller (SBC) SIP connection directly into your cloud server. We run an open-source **FreeSWITCH** container or **LiveKit SIP Gateway** to accept the SIP trunk.
Call Flow:
1. Backend triggers call via FreeSWITCH ESL (Event Socket Library) command.
2. FreeSWITCH dials customer phone number via Airtel SIP Trunk.
3. When call answers, FreeSWITCH streams 8kHz/16kHz PCM audio over WebSocket/WebRTC to Python FastAPI.
4. Python AI Agent processes speech (Deepgram STT ➔ Gemini LLM ➔ ElevenLabs/Aura TTS) and streams voice back to FreeSWITCH.

```
# FreeSWITCH ESL Call Trigger Example (Python) from ESL import * con = ESLconnection('127.0.0.1', '8021', 'ClueCon') cmd = "bgapi originate {sip_auto_answer=true}sofia/gateway/airtel/+919876543210 &socket;(127.0.0.1:8086 full)" e = con.api(cmd)
```

Approach B: Exotel Voice Stream API (India Cloud Telephony)

Architecture: Exotel exposes a cloud REST API to trigger calls and a bi-directional WebSocket Voice Stream API.
Call Flow:
1. FastAPI calls Exotel REST API `POST /v1/Accounts/{SID}/Calls/connect.json`.
2. Exotel dials customer phone number.
3. Upon connection, Exotel opens a WebSocket connection to `wss://your-domain.com/exotel/stream`.
4. Python FastAPI receives audio chunks in 8kHz mulaw format, decodes to PCM, feeds to Gemini/Deepgram, and sends audio replies back.

```
# Exotel Outbound Call Dispatch (Python FastAPI) import httpx async def trigger_exotel_call(phone_number: str, call_id: str): url = f"https://api.exotel.com/v1/Accounts/{EXOTEL_SID}/Calls/connect.json" data = { "From": phone_number, "CallerId": EXOTEL_VIRTUAL_NUMBER, "Url": f"http://your-domain.com/exotel/passthru?call_id={call_id}" } async with httpx.AsyncClient() as client: res = await client.post(url, data=data, auth=(EXOTEL_API_KEY, EXOTEL_TOKEN)) return res.json()
```

Approach C: Twilio TwiML Voice Media Stream

Architecture: Twilio Voice API using TwiML `` for bi-directional WebSockets.
Call Flow:
1. FastAPI calls `twilio_client.calls.create(to=phone_number, from_=twilio_number, url=twiml_url)`.
2. Twilio dials customer phone number.
3. When answered, Twilio fetches TwiML XML returning ``.
4. Twilio opens WebSocket and streams audio payload in base64 mulaw to FastAPI.

```
# Twilio TwiML Router Endpoint (Python FastAPI) from fastapi import APIRouter, Response @router.post("/twilio/voice") async def twilio_voice_webhook(): xml_response = ''' ''' return Response(content=xml_response, media_type="application/xml")
```

3. Summary Recommendation

1. **For Immediate Low-Complexity Indian PSTN (Best Balance)**: Implement **Exotel Voice Stream API**. It requires zero server telephony hardware, works with standard Indian phone numbers, costs **~■1.65/min total**, and plugs directly into Python FastAPI via WebSockets.
2. **For Maximum Scale & Lowest Cost**: Deploy **Airtel Corporate SIP Trunk + FreeSWITCH** in Docker. This achieves **~■1.39/min total cost** with complete infrastructure control.
3. **For International Calling**: Use **Twilio Voice API**.